

Course Code : ECT 353

GVHW/RW – 23 / 1215

**Fifth Semester B. Tech. (Electronics and Communication Engineering)
Examination**

MICROCONTROLLERS AND INTERFACING

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

- (1) All questions carry marks as indicated.
 - (2) Assume suitable data wherever necessary.
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1. Recall the RAM structure of 8051 microcontroller with a neat diagram and implement an assembly language program using looping/branching instructions to transfer 20 bytes of data from code memory location 200h onwards to internal data memory 30h onwards respectively. 7(CO1)

 2. Recall the program status word and interrupt enable register format with a diagram and implement an assembly language program to find a specific byte of data from a group of 100 bytes of data present in code memory 300h onwards and store its total count of occurrence in R0 register of bank 3. 7(CO1)

 3. Generalize the following instructions based on the number of machine cycles required, addressing mode :
 - (1) MOV C, # 01
 - (2) MOVC A, @A + DPTR
 - (3) MUL AB
 - (4) SJMP L1
 - (5) RET
 - (6) DJNZ R0, L2
 - (7) ACALL 8000 H
 - (8) RRC 8(CO1)

4. Structuring the use of timer 1 in mode 2 generate a delay of 1 second on port 1.1 pin. Please outline the question with details of calculation and description. 8(CO2)
5. Using the knowledge of ARM7TDMI please explain the register set of microcontroller. 7(CO4)
6. Elucidate the operating modes of ARM7TDMI with a detailed example. 7(CO1,4)
7. Design a display circuit where in a seven segment display (common cathode) is interfaced with ARM7TDMI and write a embedded c program to display numbers 0 to 9 continuously. 8(CO3)
8. Create a FIFO (First In First Out) algorithm for 4 tasks considering the following input data :

TASK NO	Arrival Time	Execution time
1	1	3
2	7	5
3	10	9
4	15	2

8(CO3,4)

